

Applying Newton's Laws

Worked solutions • 5.60 and 5.62

Scope: all nine even-numbered problems in the assigned list: 60, 62, 64, 74, 76, 88, 90, 92, 96. One additional odd-numbered problem is needed to reach ten.

5.60 Archaeologist on a rope

A 90.0-kg archaeologist rests at the midpoint. Each side of the rope is at angle θ below horizontal. The rope breaks above 2.50×10^4 N. Find (a) tension at $\theta = 10.0^\circ$ and (b) the minimum safe angle.

The horizontal components of the two equal tensions cancel. Their vertical components support the weight:

$$2T \sin \theta = mg$$

$$\text{(a)} \quad T = mg / (2 \sin \theta) = (90.0)(9.80) / (2 \sin 10.0^\circ)$$

Answer (a): $T = 2.54 \times 10^3$ N in each side.

(b) The smallest angle uses the largest permitted tension:

$$\begin{aligned} \theta_{\min} &= \sin^{-1}[mg / (2T_{\max})] \\ &= \sin^{-1}[(90.0)(9.80) / (2 \times 2.50 \times 10^4)] \end{aligned}$$

Answer (b): $\theta \geq 1.01^\circ$.

5.62 Two-pulley lifting system

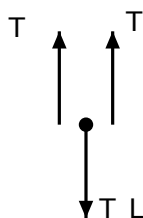
A worker pulls down on a rope to lift weight w at constant speed. Find the upper-chain tension, lower-chain tension, and applied force F . Rope, pulleys, and chains have negligible weight.

Let T be the uniform rope tension, T_L the lower-chain tension, and T_U the upper-chain tension. Each object has zero acceleration.

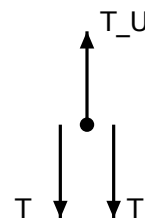
Weight



Lower pulley



Upper pulley



Weight: $T_L - w = 0$, so $T_L = w$.

Lower pulley: $2T - T_L = 0$, so $T = w/2$.

Upper pulley: $T_U - 2T = 0$, so $T_U = w$.

Answers: lower chain = w ; upper chain = w ; $F = T = w/2$.

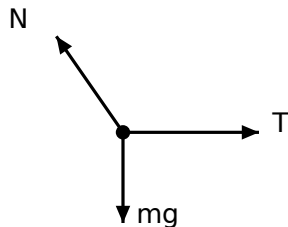
Applying Newton's Laws

Worked solutions • 5.64, 5.74 and 5.76

5.64 Ball held on a smooth ramp

A horizontal wire holds a ball of mass m on a frictionless 35.0° ramp. (a) Draw its forces. (b) Find the normal force. (c) Find the wire tension.

(a) The forces are weight mg downward, tension T horizontally right, and normal force N perpendicular to the ramp. There is no friction.



N is 35.0° from vertical.

The wire acts along a line through the center.

(b) Vertical equilibrium: $N \cos 35.0^\circ = mg$.

Answer (b): $N = mg / \cos 35.0^\circ = 1.22mg$.

(c) Horizontal equilibrium: $T = N \sin 35.0^\circ$.

Answer (c): $T = mg \tan 35.0^\circ = 0.700mg$.

5.74 Hanging mass pulls a block uphill

A 20.0-kg block is on a 53.1° incline with $\mu_k = 0.40$. Find the hanging mass m_2 required to descend 12.0 m in 3.00 s from rest.

The required acceleration is $a = 2s/t^2 = 2(12.0)/(3.00)^2 = 2.667 \text{ m/s}^2$. Friction on m_1 acts downhill because m_1 moves uphill.

$$m_1: T - m_1 g \sin \alpha - \mu_k m_1 g \cos \alpha = m_1 a$$

$$m_2: m_2 g - T = m_2 a$$

Add the equations and solve for m_2 :

$$\begin{aligned} m_2 &= m_1(a + g \sin \alpha + \mu_k g \cos \alpha) / (g - a) \\ &= 20.0[2.667 + 9.80 \sin 53.1^\circ + 0.40(9.80) \cos 53.1^\circ] \\ &\quad \div (9.80 - 2.667) \end{aligned}$$

Answer: $m_2 = 36.0 \text{ kg}$.

5.76 Static friction and a rope junction

Block A weighs 60.0 N, $\mu_s = 0.25$, and a hanging weight w acts through a three-rope junction with a 45.0° diagonal rope. Find (a) friction when $w = 12.0 \text{ N}$ and (b) maximum w for equilibrium.

At the junction: $T_d \sin 45^\circ = w$ and $T_h = T_d \cos 45^\circ$. Thus $T_h = w$. The horizontal rope pulls A right; friction acts left.

Answer (a): $f_s = 12.0 \text{ N to the left}$.

(b) $N = 60.0 \text{ N}$, so $f_{s,\max} = \mu_s N = (0.25)(60.0) = 15.0 \text{ N}$.

Answer (b): $w_{\max} = 15.0 \text{ N}$.

Applying Newton's Laws

Worked solutions • 5.88, 5.90 and 5.92

5.88 Force applied to the top block

A 2.00-kg block A rests on a 5.00-kg block B on a frictionless table. Between blocks, $\mu_s = 0.400$. A horizontal force P acts on A. Find the largest P for which the blocks move together.

Friction is the only horizontal force accelerating B. Its maximum value is:

$$f_{s,\max} = \mu_s m_A g = (0.400)(2.00)(9.80) = 7.84 \text{ N}$$

$$a_{\max} = f_{s,\max} / m_B = 7.84 / 5.00 = 1.568 \text{ m/s}^2$$

For A and B together, the friction between them is internal:

$$P_{\max} = (m_A + m_B) a_{\max} = (7.00)(1.568)$$

Answer: $P_{\max} = 11.0 \text{ N}$.

5.90 Connected blocks on frictionless slopes

A 100-kg block on a 30.0° slope is connected over a pulley to a 50-kg block on a 53.1° slope. Find (a) direction of motion, (b) acceleration, and (c) tension.

(a) Compare the downhill components of gravity:

$$100g \sin 30.0^\circ = 490 \text{ N}; 50g \sin 53.1^\circ \approx 392 \text{ N}.$$

Answer (a): the 100-kg block moves downhill; the 50-kg block moves uphill.

(b) Taking each block's direction of motion as positive:

$$100g \sin 30.0^\circ - T = 100a$$

$$T - 50g \sin 53.1^\circ = 50a$$

$$a = [100g \sin 30.0^\circ - 50g \sin 53.1^\circ] / (100 + 50)$$

Answer (b): $a = 0.654 \text{ m/s}^2$.

$$\text{(c) } T = 100g \sin 30.0^\circ - 100a = 490 - 100(0.65436)$$

Answer (c): $T = 425 \text{ N}$.

5.92 Largest hanging mass without slipping

Block B (5.00 kg) rests on A (8.00 kg). The table is frictionless, $\mu_s = 0.750$ between A and B, and A is connected over a pulley to hanging block C. Find the largest mass of C so A and B move together.

Friction accelerates B: $m_B a \leq \mu_s m_B g$. Therefore $a_{\max} = 0.750g$. Add the equations for all three blocks along their directions of motion to eliminate tension and internal friction:

$$m_C g = (m_A + m_B + m_C) a$$

$$m_C = 0.750(13.00 + m_C)$$

$$0.250m_C = 9.75$$

Answer: $m_{C,\max} = 39.0 \text{ kg}$.

Applying Newton's Laws

Worked solutions • 5.96

5.96 Two sliding blocks joined by a string

A 4.00-kg block ($\mu_k = 0.25$) is downhill from an 8.00-kg block ($\mu_k = 0.35$) on a 30.0° incline. They are connected by a string and slide downhill. Find (a) their accelerations, (b) tension, and (c) what happens if their positions are reversed.

Let $m_1 = 4.00$ kg and $m_2 = 8.00$ kg. Take downhill as positive. Both friction forces point uphill. The normal forces are $N_1 = m_1 g \cos 30^\circ$ and $N_2 = m_2 g \cos 30^\circ$.

In the original arrangement, tension pulls the lower 4-kg block uphill and the upper 8-kg block downhill:

$$m_1 g \sin 30^\circ - \mu_1 m_1 g \cos 30^\circ - T = m_1 a$$

$$m_2 g \sin 30^\circ - \mu_2 m_2 g \cos 30^\circ + T = m_2 a$$

(a) Common acceleration. Add the equations to eliminate tension:

$$\begin{aligned} a &= g \sin 30^\circ - [(\mu_1 m_1 + \mu_2 m_2)/(m_1 + m_2)] g \cos 30^\circ \\ &= 9.80 \{ \sin 30^\circ - [(0.25 \times 4.00 + 0.35 \times 8.00)/12.00] \cos 30^\circ \} \end{aligned}$$

Answer (a): both blocks accelerate at 2.21 m/s^2 downhill.

(b) String tension. Substitute into the equation for the 4-kg block:

$$\begin{aligned} T &= m_1 (g \sin 30^\circ - \mu_1 g \cos 30^\circ - a) \\ &= 4.00 [4.90 - 0.25(9.80) \cos 30^\circ - 2.21243] \end{aligned}$$

Answer (b): $T = 2.26 \text{ N}$.

(c) Reversed positions. The 4-kg block is now uphill. Because it has less friction per unit mass, it accelerates downhill faster than the 8-kg block below it. The distance between the blocks decreases, so the string goes slack. A string can pull but cannot push.

Answer (c): $T = 0$; the blocks accelerate separately while separated.

$$a_4 = g(\sin 30^\circ - 0.25 \cos 30^\circ) = 2.78 \text{ m/s}^2 \text{ downhill}$$

$$a_8 = g(\sin 30^\circ - 0.35 \cos 30^\circ) = 1.93 \text{ m/s}^2 \text{ downhill}$$

The uphill 4-kg block approaches the 8-kg block. These separate accelerations apply before any collision.

Answer check

Original order: the common acceleration (2.21 m/s^2) lies between the two free-sliding values (1.93 and 2.78 m/s^2). The string slows the 4-kg block and speeds up the 8-kg block, as the force directions require.